

EVERY DEVELOPER USES IT · FEW READ IT

What's actually in an HTTP request?

The wire format that powers the entire web · line by line.

Methods

Status Codes

Headers

Body

Versions

Real Examples

Swipe → every part, decoded.



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Message Structure

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Every HTTP message · request or response · has 4 parts.

The diagram illustrates the structure of an HTTP POST request. On the left, a sample request is shown with its components color-coded: the method and path are pink, the version is blue, headers are light blue, the CRLF separator is grey, and the JSON body is light green. On the right, a bracketed list defines these components: the **START-LINE** (method · path · version), **HEADERS** (key: value, one per line), **EMPTY LINE** (CRLF · the separator), and **BODY** (optional payload).

```
POST /api/login HTTP/1.1
Host: api.example.com
Content-Type: application/json
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXbzI1NiIsImN1b2wiOiJhZGUi
Content-Length: 53
␣ CRLF
{
  "user": "kerem",
  "pass": "*****"
}
```

- START-LINE**
method · path · version
- HEADERS**
key: value, one per line
- EMPTY LINE**
CRLF · the separator
- BODY**
optional payload

KEY TAKEAWAY

Same shape for requests and responses.

Only the start-line changes: request-line in, status-line out.

Request Line

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GET /api/users HTTP/1.1

METHOD

GET

The action to perform on the target resource.

EXAMPLE

POST /api/login

- safe · idempotent
- no body · cacheable
- GET · HEAD · OPTIONS

TARGET

/api/users

The path (request-URI)
the action is applied to.

EXAMPLE

```
/users?role=admin
```

- origin-form (path + query)
- absolute-form (proxy)
- asterisk-form (OPTIONS *)

VERSION

HTTP/1.1

The HTTP protocol version the client speaks.

EXAMPLE

HTTP/2 · keep-alive

- HTTP/1.0 · plain text
- HTTP/1.1 · keep-alive
- HTTP/2 · binary frames

KEY TAKEAWAY

Every HTTP request opens with exactly one line.

Method, where to act, and which protocol version is speaking.

Methods

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Five methods carry 99% of HTTP traffic. Each has a job.

METHOD	SAFE	IDEMPOTENT	BODY	PURPOSE
GET	✓	✓	✗	Fetch a resource.
POST	✗	✗	✓	Create or trigger an action.
PUT	✗	✓	✓	Replace a resource entirely.
PATCH	✗	✗	✓	Modify part of a resource.
DELETE	✗	✓	✗	Remove a resource.

Status Line

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HTTP/1.1 200 OK

VERSION

HTTP/1.1

Protocol version the server is responding with.

EXAMPLE

HTTP/2

- matches request
- or negotiated upgrade
- HTTP/2 → HTTP/3 (QUIC)

CODE

200

A 3-digit machine-readable result of the request.

EXAMPLE

404

- 1xx informational
- 2xx ok · 3xx redirect
- 4xx · 5xx errors

REASON

OK

A short, human-friendly description of the code.

EXAMPLE

Not Found

- informational only
- ignored by clients
- free-form text

KEY TAKEAWAY

Every response opens with its verdict.

Same three slots: version spoken, code returned, reason in plain English.

200 status	OK Standard success. Response body carries the requested resource. GET Returns the resource. POST Action succeeded · result in body. BODY Yes · the payload.
---	---

201 status	Created A new resource was created. URI returned in the Location header. AFTER POST · PUT. HDR Location: /users/42 BODY Often the new resource.
--	--

204 status	No Content Success, but nothing to send back. Useful for DELETE and silent updates. AFTER DELETE · PUT · PATCH. BODY Must be empty. HDR Headers may still carry meta.
--	---

301 status	Moved Permanently The resource has a new permanent URL. Update bookmarks and links. USE Domain change · path migration. HDR Location: https://new.example.com/ CACHE Aggressively cached.
302 status	Found Temporary redirect. The original URL still works · don't update links. USE Login flow · A/B tests. HDR Location: /login CACHE Not cached by default.
304 status	Not Modified Cached copy is still valid. Saves bandwidth by skipping the body. AFTER If-Modified-Since · If-None-Match. BODY Empty · use your cache. PERF Huge bandwidth saver.

401 status	Unauthorized No valid credentials. The server doesn't know who you are yet. MEANS Auth missing or expired. HDR WWW-Authenticate: Bearer FIX Send a valid token.
403 status	Forbidden Authenticated, but not permitted. The server knows you and still says no. MEANS Permission denied. VS 401 = who? · 403 = no. PENTEST Auth bypass target.
404 status	Not Found The resource doesn't exist, or the server pretends it doesn't. MEANS Path resolves to nothing. VS Sometimes hides a 403. BODY Often custom error page.

500 status	Internal Error Generic server-side failure. An unhandled exception, most likely. MEANS Unhandled exception. PENTEST Stack traces · info leak. FIX Check server logs.
502 status	Bad Gateway An upstream service returned garbage. The proxy gave up parsing it. LAYER Proxy · CDN · LB. COMMON Upstream crashed or timed out. PENTEST Reveals proxy chain.
504 status	Gateway Timeout Upstream took too long. The proxy stopped waiting. LAYER Proxy · gateway. DIAG Slow DB · blocking call. RETRY Often safe to retry.

Request Headers

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Six headers you see on almost every outgoing request.

Host Which virtual host the server should route to.

EXAMPLE Host: api.example.com

User-Agent Identity of the client: browser, curl, library.

EXAMPLE Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (K...

Accept Which content types the client can read.

EXAMPLE Accept: application/json, text/html

Authorization Credentials proving who is calling.

EXAMPLE Authorization: Bearer eyJhbGc...

Cookie Session state echoed back to the server.

EXAMPLE Cookie: SESSIONID=abc123; theme=dark

Content-Type How the body is encoded (POST/PUT/PATCH).

EXAMPLE Content-Type: application/json



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Response Headers

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What every server tells the client about the response.

Content-Type How the body is encoded for the client.

EXAMPLE Content-Type: application/json; charset=utf-8

Content-Length Body size in bytes; ends the response.

EXAMPLE Content-Length: 1247

Set-Cookie Server-issued session state for the browser.

EXAMPLE Set-Cookie: SESSIONID=abc; HttpOnly; Secure

Cache-Control How (and how long) it may be cached.

EXAMPLE Cache-Control: max-age=3600, public

Location Where to go next; pairs with 201/301/302.

EXAMPLE Location: /users/42

Server Server software identity. Often leaks version.

EXAMPLE Server: nginx/1.25.3



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Request Body

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Three encodings cover almost every POST you'll ever send.

JSON

Content-Type: application/json

APIs · default for modern web.

```
{
  "user": "kerem",
  "role": "admin"
}
```

Form URL-encoded

Content-Type:
application/x-www-form-urlencoded

HTML forms · OAuth token endpoints.

```
user=kerem&role=admin&active=true
```

Multipart

Content-Type: multipart/form-data;
boundary=---X

File uploads · mixed data.

```
---X
Content-Disposition: form-data; name="..."
...binary bytes...
---X--
```



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Real Example

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POST /api/login · one full request, one full response.

REQUEST → what the client sends

```
POST /api/login HTTP/1.1
Host: api.example.com
Content-Type: application/json
Authorization: Bearer eyJhbGci...
Content-Length: 53
```

```
{
  "user": "kerem",
  "pass": "*****"
}
```

RESPONSE → what the server replies

```
HTTP/1.1 200 OK
Content-Type: application/json
Set-Cookie: SESSIONID=abc; HttpOnly
Cache-Control: no-store
```

```
{ "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1b250b28iOiJ1b250b28iLCJpYXQiOiIyMDIyLTAxLTAxVjE0OjU0OjU0LjE0MjE0IiwiaWF0IjoiMTY0MjE0MjE0LjE0MjE0IiwiaXNpdCI6IjE2In0", "ttl": 3600 }
```

KEY TAKEAWAY

Same shape, opposite directions.

Request out, response in. One round-trip, two messages, four parts each.

REFERENCE WORTH KEEPING

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